

|   |         |                                                      |    |     |
|---|---------|------------------------------------------------------|----|-----|
| 4 | (a) (i) | reaction/vertical force = weight – $P \cos 60^\circ$ | C1 |     |
|   |         | $= 180 - 35 \cos 60^\circ$                           |    |     |
|   |         | $= 160 \text{ (163) N}$                              | A1 | [2] |
|   | (ii)    | work done = $35 \sin 60^\circ \times 20$             | C1 |     |
|   |         | $= 610 \text{ (606) J}$                              | A1 | [2] |

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- (b) (i) work done by force  $P$  = work done against frictional force B1 [1]
- (ii) horizontal component of  $P$  is equal and opposite to frictional force B1
- vertical component of  $P$  + normal reaction force equal and opposite to weight B1 [2]